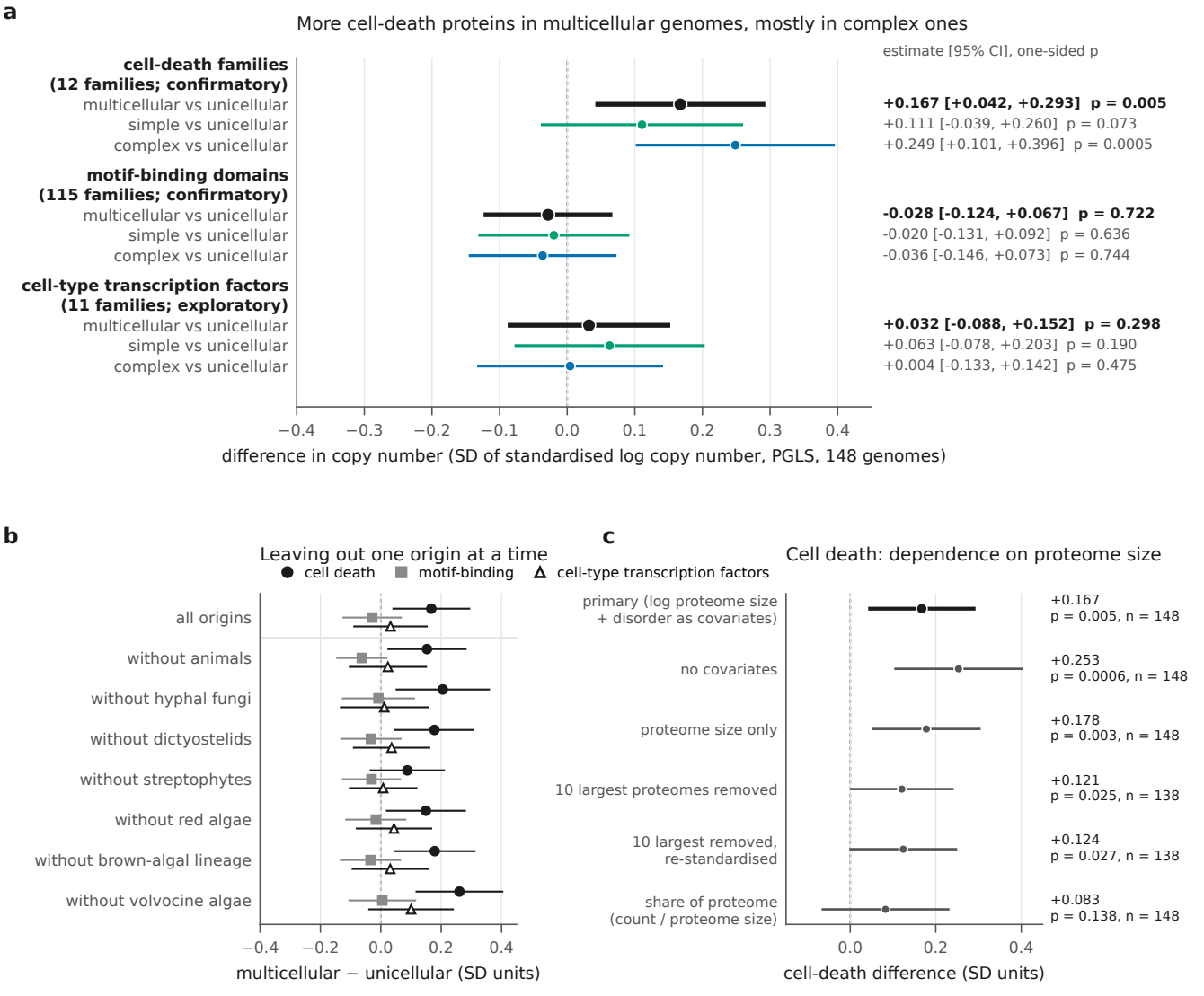
a
More cell-death proteins in multicellular genomes, mostly in complex ones
estimate [95% CI], one-sided p
cell-death families
(12 families; confirmatory)
multicellular vs unicellular
simple vs unicellular
complex vs unicellular
+0.167 [+0.042, +0.293] p = 0.005
+0.111 [-0.039, +0.260] p = 0.073
+0.249 [+0.101, +0.396] p = 0.0005
motif-binding domains
(115 families; confirmatory)
multicellular vs unicellular
simple vs unicellular
complex vs unicellular
-0.028 [-0.124, +0.067] p = 0.722
-0.020 [-0.131, +0.092] p = 0.636
-0.036 [-0.146, +0.073] p = 0.744
cell-type transcription factors
(11 families; exploratory)
multicellular vs unicellular
simple vs unicellular
complex vs unicellular
+0.032 [-0.088, +0.152] p = 0.298
+0.063 [-0.078, +0.203] p = 0.190
+0.004 [-0.133, +0.142] p = 0.475
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4
difference in copy number (SD of standardised log copy number, PGLS, 148 genomes)
b
Leaving out one origin at a time
cell death
motif-binding
cell-type transcription factors
all origins
without animals
without hyphal fungi
without dictyostelids
without streptophytes
without red algae
without brown-algal lineage
without volvocine algae
−0.4 −0.2 0.0 0.2 0.4
multicellular − unicellular (SD units)
c
Cell death: dependence on proteome size
primary (log proteome size
+ disorder as covariates)
no covariates
proteome size only
10 largest proteomes removed
10 largest removed,
re-standardised
share of proteome
(count / proteome size)
+0.167
p = 0.005, n = 148
+0.253
p = 0.0006, n = 148
+0.178
p = 0.003, n = 148
+0.121
p = 0.025, n = 138
+0.124
p = 0.027, n = 138
+0.083
p = 0.138, n = 148
0.0 0.2 0.4
cell-death difference (SD units)